

Hardness of computation of quantum invariants on 3-manifolds with restricted topology

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Theorem [Alagic & Lo, 2014]:

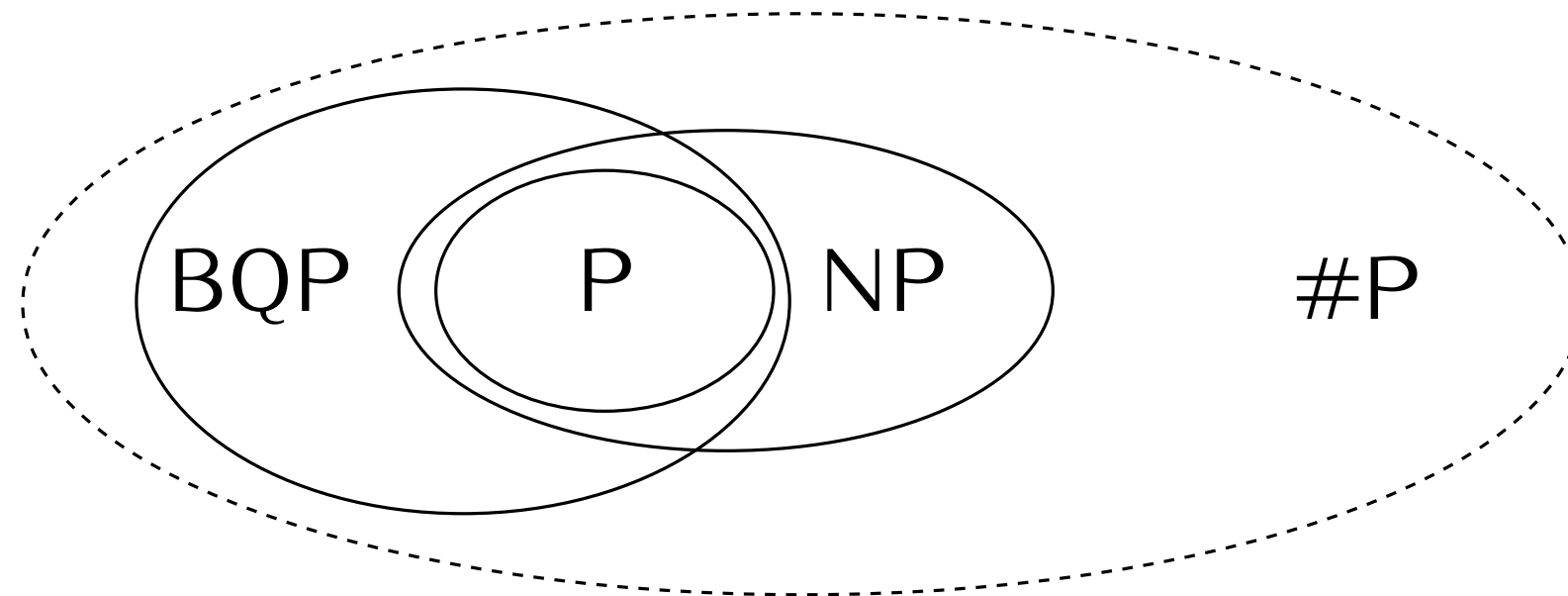
Computing (or even well-approximating) the Reshetikhin–Turaev invariant for general closed 3-manifolds is #P-hard.

The Reshetikhin–Turaev invariant

Considering a logic sentence with free variables

$$T : (x \wedge \neg y) \vee z$$

- P: given an assignment, telling whether the assignment gives a TRUE statement.
- NP-hard: for a given sentence, telling whether there exists **one** assignment that makes the sentence TRUE.
- #P-hard: for a given term, **counting** how many assignments make the sentence TRUE.



Our contribution

Does restricting the topology of M make the problem easier?

- **irreducible*** if it is not homomorphic to a direct sum $N_1 \# N_2$ where $N_1, N_2 \neq S^3$;
- **hyperbolic** if it can be equipped with a hyperbolic metric;
- **small*** if every small orientable embedded surface is compressible.

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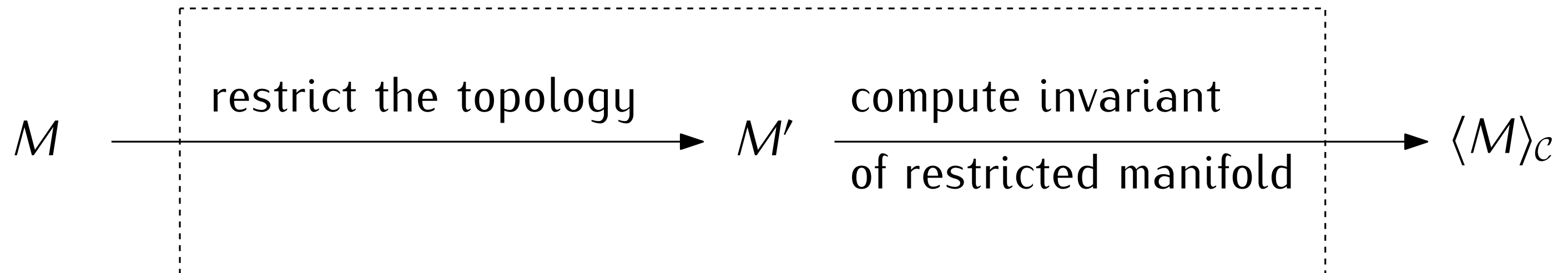
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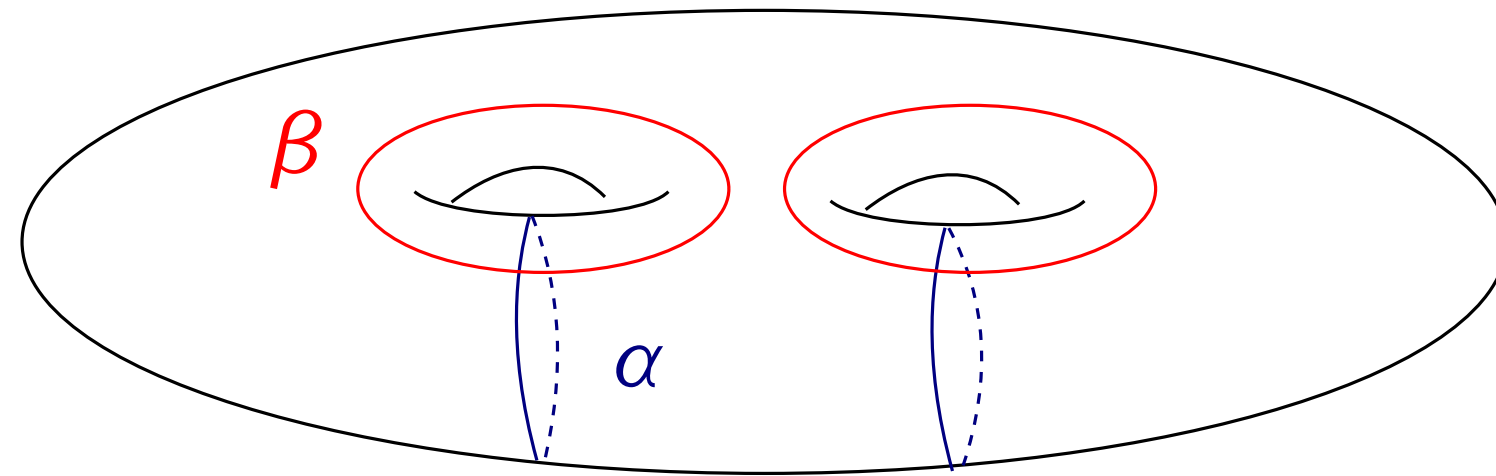
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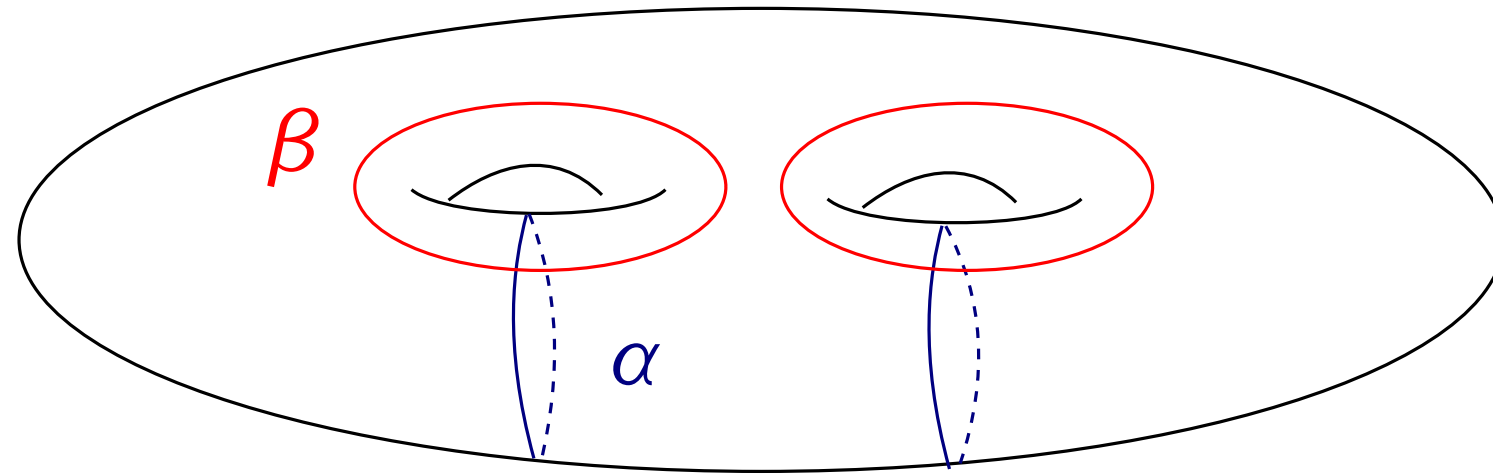
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- A **Heegaard diagram** represents a closed 3-manifold with sets of curves on a surface.



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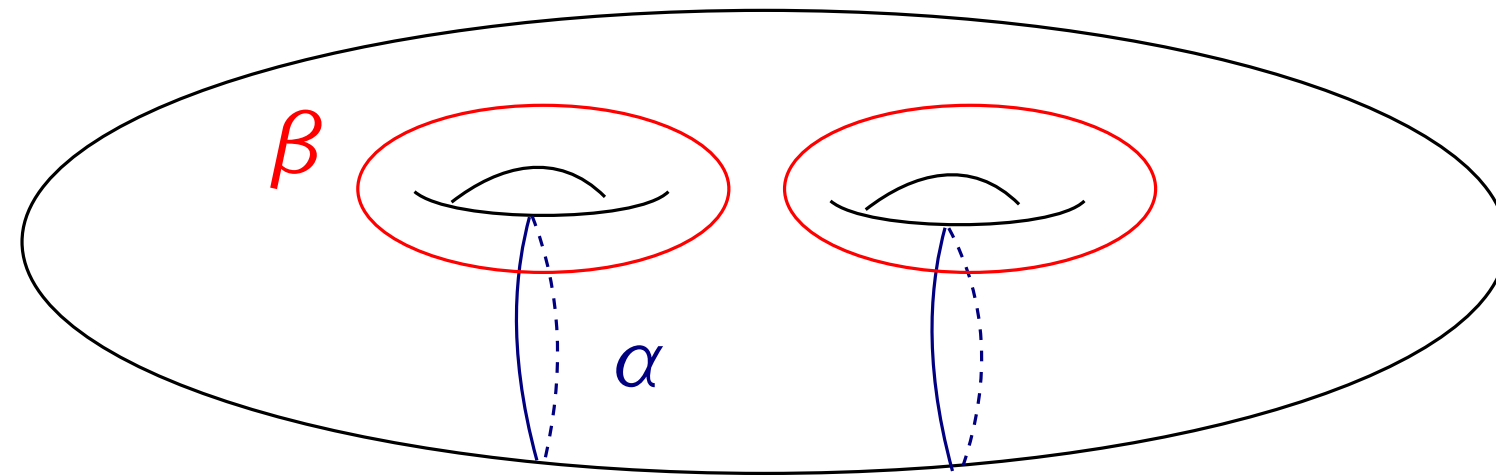
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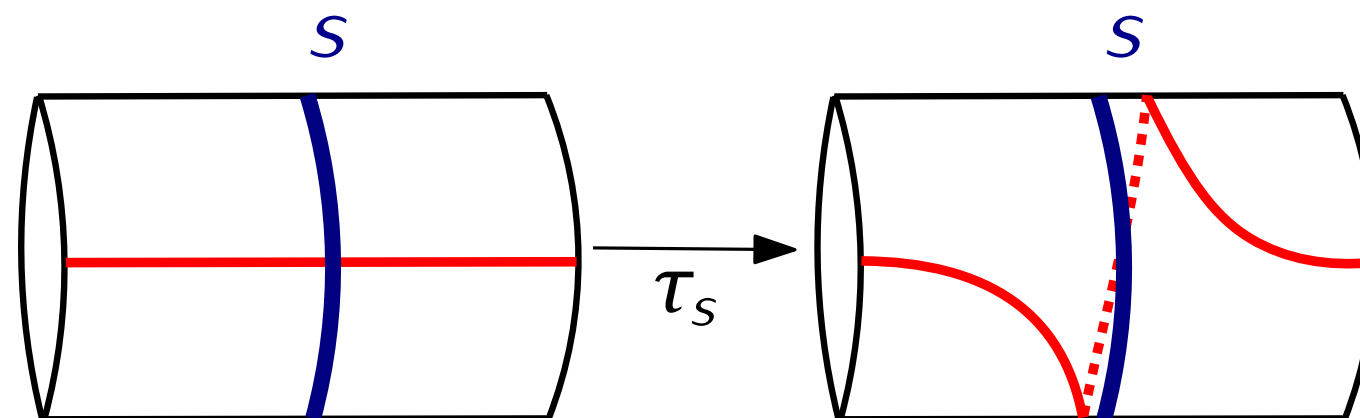
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- We can use **Dehn twists** about s to complicate the diagram.



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Theorem [Vafa, 1988]:

For every choice of $\mathcal{C}$, there is a $N \in \mathbb{Z}$ such such that τ_s^{kN} does not alter the invariant $\langle \cdot \rangle_{\mathcal{C}}$.

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Theorem:

For (Σ, α, β) representing M , there is a $O(\text{poly})$ algorithm that outputs $(\Sigma', \alpha', \beta')$ with larger Hempel distance, but representing M' with the same invariant.

Dziękuję bardzo!